



Spatial and temporal receptive fields of geniculate and cortical cells and directional selectivity

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Abstract

The spatio-temporal receptive fields (RFs) of cells in the macaque monkey lateral geniculate nucleus (LGN) and striate cortex (V1) have been examined and two distinct sub-populations of *non-directional* V1 cells have been found: those with a slow largely monophasic temporal RF, and those with a fast very biphasic temporal response. These two sub-populations are in temporal quadrature, the fast biphasic cells crossing over from one response phase to the reverse just as the slow monophasic cells reach their peak response. The two sub-populations also differ in the spatial phases of their RFs. A principal components analysis of the spatio-temporal RFs of *directional* V1 cells shows that their RFs could be constructed by a linear combination of two components, one of which has the temporal and spatial characteristics of a fast biphasic cell, and the other the temporal and spatial characteristics of a slow monophasic cell. Magnocellular LGN cells are fast and biphasic and lead the fast-biphasic V1 subpopulation by 7 ms; parvocellular LGN cells are slow and largely monophasic and lead the slow monophasic V1 sub-population by 12 ms. We suggest that directional V1 cells get inputs in the approximate temporal and spatial quadrature required for motion detection by combining signals from the two non-directional cortical sub-populations which have been identified, and that these sub-populations have their origins in magno and parvo LGN cells, respectively. © 2000 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Many striate cortex simple cells have directional selectivity, responding robustly to a pattern of the optimal spatial frequency and orientation moving in one direction but little if at all to a stationary pattern or to a pattern moving in the opposite direction (Hubel & Wiesel, 1962; Emerson & Gerstein, 1977; De Valois, Yund, & Hepler, 1982; Reid, Soodak, & Shapley, 1987; Albrecht & Geisler, 1991). Such directional selective cells are presumed to constitute the first stage in a system to detect and characterize the motion of visual objects. By definition, a moving object is at one location at a given time and in a different location at a later

time, and could be identified as such by a cell that sums inputs that differ from each other appropriately in *both* space and time. Here the question of what those inputs might be in the case of directionally-selective cells in macaque monkey striate cortex is addressed.

Adelson and Bergen (1985) and Watson and Ahumada (1985) have proposed widely accepted motion models which the data presented here strongly support. Both are multistage models, with properties at successive stages that might well correspond to the characteristics of simple and complex V1 cells, respectively. Here one is concerned with the possible inputs to the 1st stage.

The inputs to directional cells in each of these motion models have three important characteristics: (a) the local input images have been bandpass filtered in spatial frequency and orientation. Such filtering would vastly simplify the correspondence problem, that of

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determining what stimulus in temporal interval 2 corresponds to the stimulus in temporal interval 1; (b) there must be two (or more) inputs that differ from each other in spatial location or in spatial phase (ideally in spatial quadrature); and (c) these two inputs must also differ from each other in latency or in temporal phase (ideally in temporal quadrature).

It has been accepted for some years that the magnocellular (M_c) lateral geniculate nucleus (LGN) cells provide the input to the motion system (Livingstone & Hubel, 1987, 1988; Maunsell, 1987). This is based on considerable anatomical, physiological and psychophysical evidence. Anatomical studies have shown a path from M_c cells to cells in striate cortex layer IV α , then to layer IVb, and from there to MT (V5), an area which is clearly involved in motion processing (Zeki, 1974; Albright, 1984; Movshon, Adelson, Gizzi, & Newsome, 1985). Furthermore, lesions in the magno LGN layers produce a loss of motion sensitivity, more so than do parvocellular (P_c) lesions (Merigan & Maunsell, 1990; Schiller, Logothetis, & Charles, 1990). Physiological studies (Kaplan & Shapley, 1982) have shown that M_c cells have a high contrast-gain and often saturate at about 10% contrast; these properties resemble those of the perceptual motion system, which is characterized by high contrast sensitivity with performance which often does not improve with contrasts above about 10% (Nakayama & Silverman, 1985, but see Edwards, Badcock, & Nishida, 1996).

These studies all provide strong support for M_c cells having a major input to the motion system, but they do not preclude a P_c input as well. For instance, there could be input to cells in layers IV α and IVb of the magno path, from units in other cortical layers that have a parvo input. Furthermore, studies that showed major deficits in activation of MT cells from magno LGN lesions also found significant deficits from lesions in the parvo layers as well (Nealey & Maunsell, 1994). Most tellingly, the arguments for M_c cells being the sole basis for motion sensitivity have ignored a basic requirement for motion detection, namely inputs that differ from each other in latency or temporal phase (as well as in location or spatial phase). Our recordings from LGN cells confirm an earlier report (Schmolesky, Wang, Hanes, Thompson, Leutgeb, Schall, & Leventhal, 1998) that magno cells all have virtually identical latencies and thus could not provide the variable latencies needed for a motion system without an elaborate intracortical delay arrangement for which there is no anatomical or physiological evidence.

To examine this question further, we have studied, in single-cell recordings, the spatial and temporal receptive field (RF) properties of both directional and non-directional striate cortex cells, and also of different classes of LGN cells. The presence of two subpopulations of non-directional striate cortex cells are reported here

that have precisely the spatial and temporal phase differences needed as inputs to the directionally-selective cells that have been recorded, to account for their directional selectivity. Furthermore, these two classes of non-directional cells appear from the results to have inputs from M_c and P_c LGN cells, respectively. This suggests that the visual system gets the required latency difference (and perhaps also the spatial difference) for motion detection by combining inputs that have their origin in fast biphasic M_c cells and slow more monophasic P_c cells, rather than just from M_c cells as has been thought.

A preliminary account of some of these results on the temporal RFs of V1 cells has been published (De Valois & Cottaris, 1998). Here a more extensive cortical cell sample is reported on in which not just the temporal RFs of cortical simple cells have been analyzed, as before, but also their spatial RFs, and the spatio-temporal RFs of P_c and M_c LGN cells as well.

2. Methods

The data presented here are from single-cell recording experiments carried out on anesthetized, paralyzed macaque monkeys (*Macaca mulatta* and *Macaca fascicularis*). The animal was first tranquilized with ketamine HCl (10–15 mg/kg, i.m.) and then anesthetized with a continuous i.v. infusion of sufentanil citrate (8–12 μ g/kg per h during surgery; 5–8 μ g/kg per h during recording). To avoid pain from ear and eye bars, a stainless steel pedestal was bolted to the skull during surgery while the animal was held in the stereotaxic plane. The head was then held painlessly by this pedestal for the duration of the experiment. For cortical recordings, a small hole was bored through the skull, the dura mater thinned, and the hole filled with agar and bone wax after insertion of the microelectrode. For LGN recordings, a 1 cm hole centered at stereotaxic coordinates F 7, L 11 was drilled through the skull, the dura reflected, and the hole again sealed with agar and bone wax after insertion of the microelectrode. Small marking lesions were made along the recording tracks, and at the termination of the experiment the brain was blocked and sectioned in the electrode plane and the recording sites determined from histological examination of the electrode tracks and marking lesions.

After surgery, paralysis was induced and maintained with pancuronium bromide (0.1–0.15 mg/kg per h, i.v.). The electrocardiogram, the electroencephalogram, the expired CO₂ and body temperature were continuously monitored and maintained within normal limits. All of the procedures were in accord with National Institutes of Health guidelines and were approved by the local Animal Care and Use Committee.

Visual stimuli were generated and controlled by a Sun/TAAC computer graphics system and presented on a NEC color monitor with a spatial resolution of 1024×900 pixels, a 66-Hz refresh rate and 70 cd/m^2 mean luminance. The stimuli consisted of 50% luminance contrast increments and decrements from this mean level. Spikes were recorded with tungsten microelectrodes from single cells in the LGN and striate cortex with a 1 ms resolution, and were analyzed on-line by the Sun workstation. The recording sites were determined from histological examination of the electrode tracks and marking lesions.

3. Results

The concept of a RF was introduced to vision by Hartline (1940), in his studies of single fibers in *Limulus*. The RFs of visual neurons have proved a useful tool for understanding the function of the system at early levels, up through simple striate cortex neurons. Its usefulness depends upon the system being reasonably linear up to the level studied. Insofar as that is true, a spatial RF based on the responses to a spot in different locations will allow one to predict the cell's responses to other more interesting stimuli. Complex cells, and cells at still higher levels, have major non-linearities that make the usual RF maps not very useful (Shapley, 2000).

However, the conventional RF mapping procedure, even when used to examine cells early in the system, has major limitations. One is that it gives just a spatial map, ignoring the vitally important temporal dimension. The other is that a single spot or bar is not a very effective stimulus for many cells, so weaker portions of the RF may not be seen at all. Two developments have largely overcome each of these limitations. The reverse correlation procedure (de Boer & Kuypers, 1968; MacLean, Raab, & Palmer, 1994) allows one to measure the fine temporal evolution of the neuron's responses at each of a number of spatial locations. This white-noise stimulation method presents small, bar-shaped stimuli in rapid succession and at random spatial locations. In this way, neurons are stimulated with input signals of low spatio-temporal energy and are more likely to operate in their linear input–output regime, thus avoiding the large transient (and possibly non-linear) response components evoked by the prolonged presentation of traditional on/off stimulus presentations. The method is also very efficient since it allows presentation of a large number of stimuli in rather small time intervals. Furthermore, computation of the event-triggered stimulus-response cross-correlation function is much faster in the reversed direction than in the forward direction because trigger events are the evoked spikes which are generally fewer than pre-

sented stimuli, which are the trigger events in the forward direction.

A second important development in RF procedures is the use of multiple stimuli presented simultaneously and modulated independently of each other. This is achieved by using shifted versions of binary m-sequences as the controlling signals (Sutter, 1992; Reid, Victor, & Shapley, 1997; Anzai, Ohzawa, & Freeman, 1999). Multiple stimulus presentation allows testing of neurons with a richer and more powerful set of stimulus configurations than does the single stimulus presentation procedure.

In the reverse correlation RF mapping procedure, one presents a long continuous series of stimuli while simultaneously recording the spike responses of a cell. Each time the cell fires a spike, the computer determines what stimuli at each spatial location had been presented at each of various times preceding the spike. In the course of a mapping a single RF, some thousands of spikes might be recorded, resulting in a spatio-temporal RF map showing the correlation between the spike discharge and the stimuli at different spatial locations and at different times before the spikes. Given the fact that there is a significant latency between the presentation of a stimulus and the arrival of information at the geniculate or cortex, the correlation of the spike discharge with the particular stimuli presented in each location would be essentially zero for intervals shortly before each spike. However, after longer intervals between presentation of stimuli and the spike, positive (and negative) correlations would be seen between the presentation of certain stimuli and the occurrence of a spike. Reverse correlation was typically carried out in 5 ms steps back to 200 ms before the spike, allowing one to see not just the latency of the response, but how it changed over time, e.g. whether the response built up and then tapered off in a monophasic time course or whether it reversed phase at some point to produce a biphasic temporal impulse function. The resulting combined spatio-temporal RF also allows one to see whether the spatial RF is constant over time, or whether it shifts in spatial position with time, as would be true for a directional cell in a linear motion system.

The second aspect of the RF mapping procedure was the simultaneous presentation of multiple RF-mapping stimuli, rather than just one stimulus at a time. In a conventional RF mapping procedure with single stimuli presented in each of a number of different locations, each stimulus covers perhaps 1/10 of the RF and thus evokes a relatively small response. As a consequence, one may not see weaker sidebands within the RF, but these sidebands nonetheless make a significant contribution to the tuning properties of a cell to more extensive patterns. (This is one reason that cortical cells often have narrower spatial frequency and orientation

tuning (Tadmor & Tolhurst, 1989) than one would predict from a Fourier transform of their spatial RFs as conventionally mapped.) With the simultaneous presentation of stimuli at each location, appropriate stimuli to the sidebands would on occasion occur together with optimal stimuli to the main RF regions, and the contributions of the sidebands would be noted in the increased probability of a spike's occurring.

To gather the data reported here, M-sequence reverse correlation was carried out with 12–16 bars of the optimal orientation for the cell. In each 30-ms interval, each bar would be either a luminance increment or decrement with respect to the mean, generally of 50% Michelson contrast. A single stimulus would thus be a pattern of 12–16 black and white bars, followed the next 30 ms by a different pattern of black and white bars, etc. the sequence continuing for several minutes. One of the advantages of using an M-sequence presentation is that it allows one to examine non-linear as well as linear components of the responses. However, both LGN and V1 simple cells are quite linear in their summation properties, and only the linear kernels are reported on here.

3.1. Directional and non-directional striate cells

In initial tests on each V1 cell, the responses to drifting grating patterns of the optimal spatial frequency were recorded at each of various orientations. Striate cortex cells typically have a preferred orientation, but they vary widely in the extent to which they respond to a grating of that orientation drifting in one

direction versus the opposite direction. To quantify a cell's degree of directional selectivity, we define a

$$\text{Direction Index} = (1 - (np/p)) \times 100 \quad (1)$$

where p is the response in the preferred direction, and np is the response in the non-preferred direction. The response measure in each case was the amplitude of the fundamental component in a Fourier analysis of the cell's post-stimulus-time-histogram in response to drifting sine-wave gratings. A score of 100 would indicate a cell that responded to a pattern moving in one direction but not at all to that pattern moving in the opposite direction, and 0 would indicate a cell that gave exactly the same response to movement of the optimal orientation grating in either direction. As was reported earlier (De Valois et al., 1982), most V1 cells have some degree of directional preference, but a plot of the directional indices of a population of V1 cells reveals a distinctly bimodal distribution, with a small group of very directional cells. The data from the current study confirm the presence of such a distinct subgroup of very directional cells, see Fig. 1. Only those cells with a direction index of more than 70 were classified as being directional. Of the 171 striate cortex simple cells studied, 134 were classified as non-directional and 37 as directionally selective.

3.2. Spatial and temporal phase

Fig. 2 shows typical spatio-temporal RFs of a sample of V1 simple non-directional cells. Shown are contour plots, in which the spatio-temporal regions giving excitatory responses to increments in light are shown in red and those regions exhibiting excitation to decrements are shown in blue. Each of these cells responds to stimuli over a limited region in space (x axis) and a limited region in time (y axis). What one is concerned with here are shapes of the spatial and temporal response waveforms. The response shapes will be characterized in terms of their symmetry or their phase with respect to the center of the response in both space and time. Thus a horizontal cross-section through the RF of the cell shown in Fig. 2a would show that its spatial RF to be triphasic and even-symmetric with respect to the RF center. One alternatively can refer to this RF as being in cosine spatial phase. On the other hand, the cell whose RF is shown in Fig. 2c has an odd-symmetric or sine spatial RF.

A vertical cross-section through the RF of the cell shown in Fig. 2a would show that its temporal RF is monophasic and thus even-symmetric with respect to the center of the response period, the response just building up and then dying off symmetrically in time. This cell can thus be characterized as having an even-symmetric temporal as well as spatial RF. On the other hand, the cell illustrated in Fig. 2e has a temporal RF

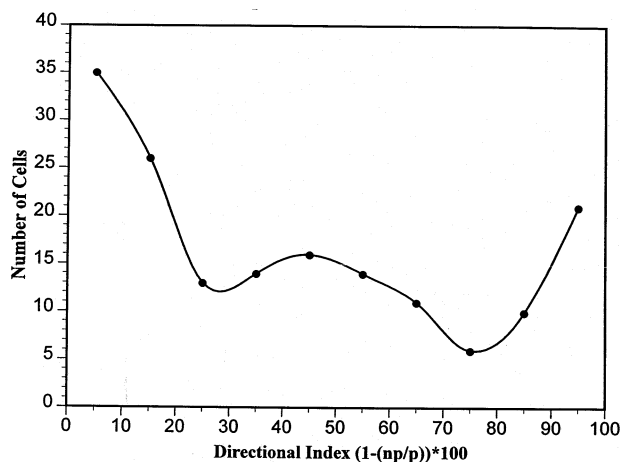


Fig. 1. The distribution of directional selectivity for the population of 171 striate cortex simple cells, computed from the cells' responses to optimum spatial frequency gratings drifted in opposite directions at the optimal orientation. A direction index of 0 indicates identical responses to the two directions and thus total lack of directional selectivity; an index of 100 indicates a response to one direction of motion but no response to the opposite direction and thus total directional selectivity. Cells with direction indices of 70 or more were classified as directional units.

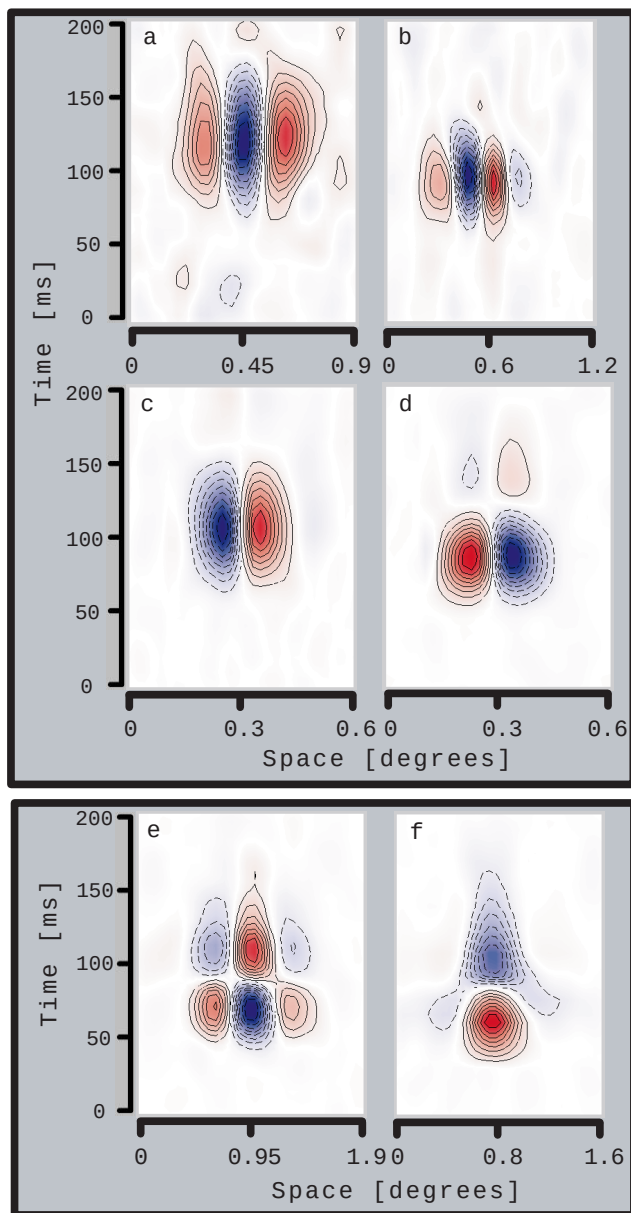


Fig. 2. Examples of the spatio-temporal receptive fields (RFs) of a sample of V1 simple cells, obtained from reverse-correlation mapping with m-sequence presentation of multiple bars. Increasing saturation of red indicates increasing correlation between the occurrence of a spike and the presentation of a light increment at that spatial location and that time before the spike; various saturations of blue indicate the degree of correlation between firing and the presentation of light decrements at that spatio-temporal location. (a–d) The RFs for four monophasic cells. Note that they are not only monophasic in their temporal responses, but they have relatively long latencies and a variety of different spatial phases to their RFs, approximately cosine (cell a), sine (cells c & d, but with spatial phases shifted 180° with respect to each other), and intermediate (cell b). (e and f) The spatio-temporal RFs of two biphasic cells. Note not only that they have a biphasic temporal response, but that they have short latencies and both are in cosine spatial phase, one with RF center response to decrements and one with center responsive to increments. In the population as a whole, the cross-points in the temporal RFs of these cells coincide with the peaks of the responses of the monophasic cells.

which is biphasic and thus odd-symmetric with respect to the center of the response period. It is to be emphasized that one is concerned here with the shapes of the spatial and temporal waveforms, not the phase relation between stimulus and response.

3.3. Temporal characteristics of non-directional V1 cells

Examining numerous samples of the RFs of non-directional striate cortex simple cells, we were struck by the fact that whereas most cells have a largely or totally monophasic temporal impulse response (e.g. Fig. 2a–d), an occasional cell was found to have a very biphasic temporal RF (e.g. Fig. 2e,f). To quantify the temporal phasic characteristics of the cells, the amplitude of the initial temporal phase of the RF, and of the second, reversed phase (if present) for each non-directional simple cell in this sample was measured. We define a

$$\text{Biphasic Index} = \frac{\text{amplitude of 2nd temporal peak}}{\text{amplitude of 1st peak}} \quad (2)$$

A high biphasic index thus indicates a very biphasic cell and a low index mostly, or entirely, monophasic, cell. Fig. 3 shows the distribution of biphasic indices for all the non-directional cells in this sample. It can be seen that most V1 cells are largely monophasic in their temporal response. In the case of cells with small biphasic indices, the response just builds up and then gradually dies off in time. However, the cells in a smaller sub-population show a quite biphasic temporal impulse response, the initial response dropping sharply

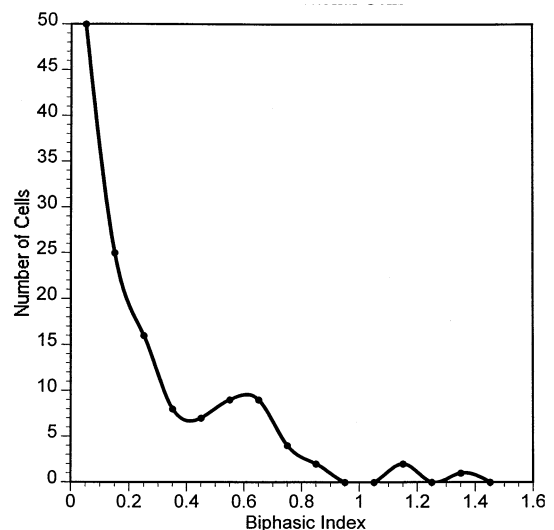


Fig. 3. The distribution of the biphasic index of the population of non-directional V1 simple cells. Note that most of the simple cell population has a quite monophasic temporal impulse response (small biphasic index), a distinct sub-population is quite biphasic. Cells with a biphasic index of 0.5 or more (reversed phase half as large or more than the initial phase) were classified as biphasic cells.

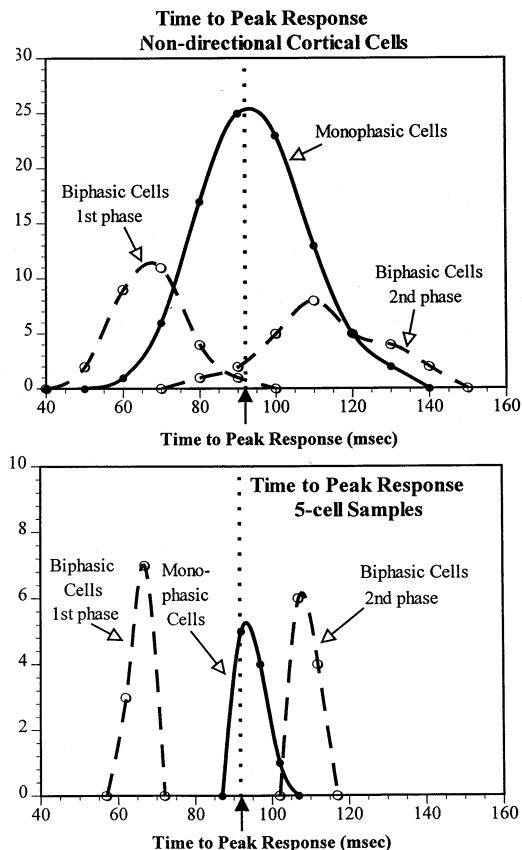


Fig. 4. (a) The distribution of time to peak response, calculated from the spatio-temporal receptive fields (RFs), of the population of monophasic cells, and of both the initial and the later reversed phase of the biphasic cells. Note that the monophasic cells reach their peak response at almost exactly the time that the biphasic cells are reversing phase, shown by the average cross-over point (vertical dashed line and arrow). The two populations are thus in temporal quadrature. (Note that the cross-over point of the biphasic cells is slightly past the midpoint between the two response peaks, because the initial response phase is larger than the reversed phase.) (b) The distribution of peak response times of ten samples of five cells each, chosen at random from the populations of biphasic cells and of monophasic cells, respectively. It can be seen averaging across as few as five cells from each of these response types would result in non-overlapping distributions of peak response times for the biphasic and monophasic cells. Random samples from these two populations would thus almost always be in temporal quadrature.

and then reversing in polarity. The distribution of biphasic indices of the cells shown in Fig. 3 is clearly bimodal, and in fact is well fit by the sum of an exponential, for the largely monophasic cells, and a Gaussian, for the biphasic ones.

The data shown in Fig. 3 indicate that there are two distinct sub-populations of non-directional cells in V1, in terms of the phasic characteristics of their temporal responses. To further examine other properties of each of these two cell types, the cells were divided into two classes, purely on the basis of the biphasic index: largely

monophasic cells, with a biphasic index of less than 0.3, and strongly biphasic cells with a biphasic index of 0.5 or more (i.e. with the second response phase more than half as large as the first). For convenience, these will be referred to as monophasic and biphasic cells, respectively. Those 15 cells with biphasic indices of 0.3–0.49 were considered ambiguous and were not further considered (it was found that grouping these few ambiguous cells with either the monophasic or the biphasic populations, or splitting the populations at 0.4, makes no significant difference to any of the conclusions). The non-directional simple cortical cell sample thus consisted of 92 monophasic and 27 biphasic cells. The biphasic indices for the four largely monophasic cells shown in Fig. 2a–d are respectively 0, 0, 0.1, and 0.14; for the two biphasic cells shown there (Fig. 2e,f), the biphasic indices are respectively 0.69 and 0.56.

3.4. Latencies of monophasic and biphasic non-directional cells

It was clear to the authors from examining the spatio-temporal RFs that the temporally-biphasic cells typically had a considerably shorter response latency than the monophasic cells. Although the delay to response onset is the usual measure of latency, of more interest is the time to the peak of the responses. This can often be more precisely specified, but more importantly it allows one to examine the temporal properties of both components of the responses of biphasic cells and to compare these with the time-to-peak of the monophasic population. This is of principal interest with respect to the issue raised below of the relationship between these two non-directional cell populations and directionally selective striate cortex simple cells.

Fig. 4 shows the distribution of peak response latencies of the monophasic and the biphasic non-directional cell populations. It can be seen that the biphasic cells on average are much faster than the monophasic cells, reaching their peak responses about 25 ms sooner (median latency of 68 vs. 93 ms for the monophasic population). The biphasic cells, by definition, then reverse in phase and on average reach their second peak at about 112 ms. This is illustrated by the cells whose RFs are shown in Fig. 2. The times-to-peak of the four monophasic cells (Fig. 2a–d) are respectively 114, 94, 102, and 84 ms, respectively. The two biphasic cells shown (Fig. 2e,f) have times-to-peak of 69 and 60 ms for the initial phase of their response, and of 106 and 101 ms, respectively, for the second reversed phase of their responses.

Note, in Fig. 4, that the response time course of these two non-directional cell populations are approximately 90° shifted in phase relative to each other, that is, they are on average in close to temporal quadrature. That is, the population of biphasic cells is shifting over from one phase to the reverse (shown in Fig. 4 by the

average cross-over point) at the time that the monophasic cell population reaches its peak response.

It is apparent in Fig. 4a that, while the two cell populations are in near temporal quadrature with respect to each other, there is considerable overlap in the distributions. Because of this variability, pairs of random single samples from these two distributions, just one biphasic and one monophasic cell, would often not be in quadrature phase. However, if as few as five samples from each of the two populations were averaged together, the biphasic and monophasic cell distributions would be quite discrete and the two populations would almost always be within a few ms of precise temporal quadrature, see Fig. 4b.

3.5. Spatial phases of monophasic and biphasic non-directional cells

From the spatio-temporal RF measurements for luminance-varying patterns, one can examine the spatial and temporal characteristics of the cells' RFs. As has been shown for simple striate cortex cells in cat (Kulikowski & Bishop, 1981; Webster & De Valois, 1985; Field & Tolhurst, 1986; Jones & Palmer, 1987), a cross-section through the spatial RF of a non-directional cell orthogonal to its preferred orientation can be fit reasonably well with a Gabor function, i.e. a sinusoid tapered by a Gaussian. Of interest here is the spatial phase of the best-fitting sinusoid for each of the cells in the sample, in particular whether there are differences in the distribution of spatial phases between

the cells in the two sub-populations identified on the basis of their temporal properties.

To quantify the spatial phase of a cell's RF, the best-fitting Gabor to a cross-section through the RF was found at the peak response time. The fitting function determined the optimal location, amplitude and standard deviation of the Gaussian, and the optimal amplitude and phase of the sinusoid. The cell's spatial phase was defined as being the phase of a cosine with respect to the RF center, the center of the Gaussian. Thus a cell with an even-symmetric RF with the center region excitatory to increments would have a spatial phase of 0° , and a cell with a spatial phase of 180° would have an even-symmetric RF with a center excitatory to decrements. These even-symmetric RFs are referred to as being in cosine phase. A cell with an odd-symmetric, or sine phase, RF with a spatial phase of 90° would have a sub-region excitatory to light increments to the right of the RF center and one excitatory to light decrements to the left of center; a cell with a spatial phase of 270° would have the converse arrangement.

It is to be noted that retinal ganglion cells and LGN cells, with their radially-symmetric center-surround organization, all have approximately even-symmetric (cosine) RFs with centers excitatory to either increments or decrements; that is, spatial phases of either 0 or 180° , respectively, by our definition. The spatial phases of cortical cell RFs, on the other hand, vary from cell to cell. Hubel and Wiesel (1962) showed diagrams of the RFs of two V1 cells, one of which had what is being termed a cosine and one a sine spatial RF. From this, some investigators concluded that there were just these two spatial cell types, and referred to the cosine- and sine-RF cells as bar and edge detectors, respectively. However, quantitative studies since have shown that V1 cells of not just sine and cosine, but of all intermediate RF spatial phases are to be found (Hamilton, Albrecht, & Geisler, 1989).

It was found that the temporally-monophasic V1 cells (which constitute the majority of the V1 population) show great variability in their spatial RFs, with an approximately uniform distribution of optimal spatial phases across the population. This is illustrated by the four temporally-monophasic cells shown in Fig. 2a–d, whose respective spatial phases are 170 , 105 , 95 , and 265° . On the other hand, it was found that virtually all of the fast biphasic striate cortex cells have cosine phase RFs, with remarkable similarity across the population. The two temporally-biphasic cells whose RFs are shown in Fig. 2e,f, for example, have spatial phases of 190 and 5° , respectively. In Fig. 5 the distributions of the percentage of cells in each of these populations that have RFs of these various spatial phases are shown. Cells having RFs of the same shape but different polarities are combined in this figure (e.g. 90°

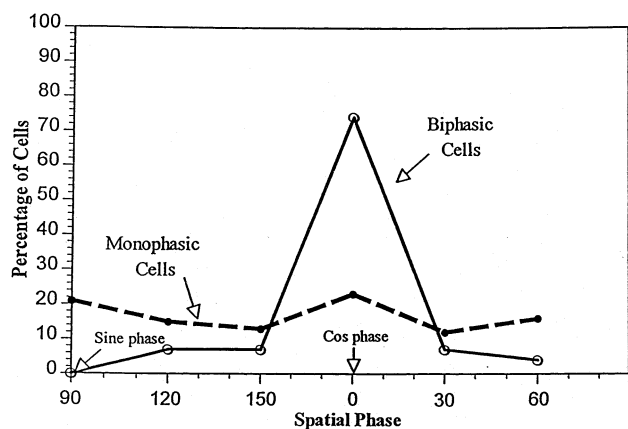


Fig. 5. The distributions of spatial phases for the populations of monophasic and biphasic cells, respectively. Shown are the percentages of cells of these two temporal classes whose receptive fields (RFs) are in various spatial phases. The spatial phases have been folded over so that cells with the same RF shape but different polarity of response are combined (e.g. all cells in cosine phase, such as the biphasic cells shown in Fig. 2E&F, are combined: $0^\circ = 0^\circ + 180^\circ$). Note that while the monophasic cells have spatial RFs of every shape, biphasic cells almost all have cosine (evensymmetric) spatial RFs.

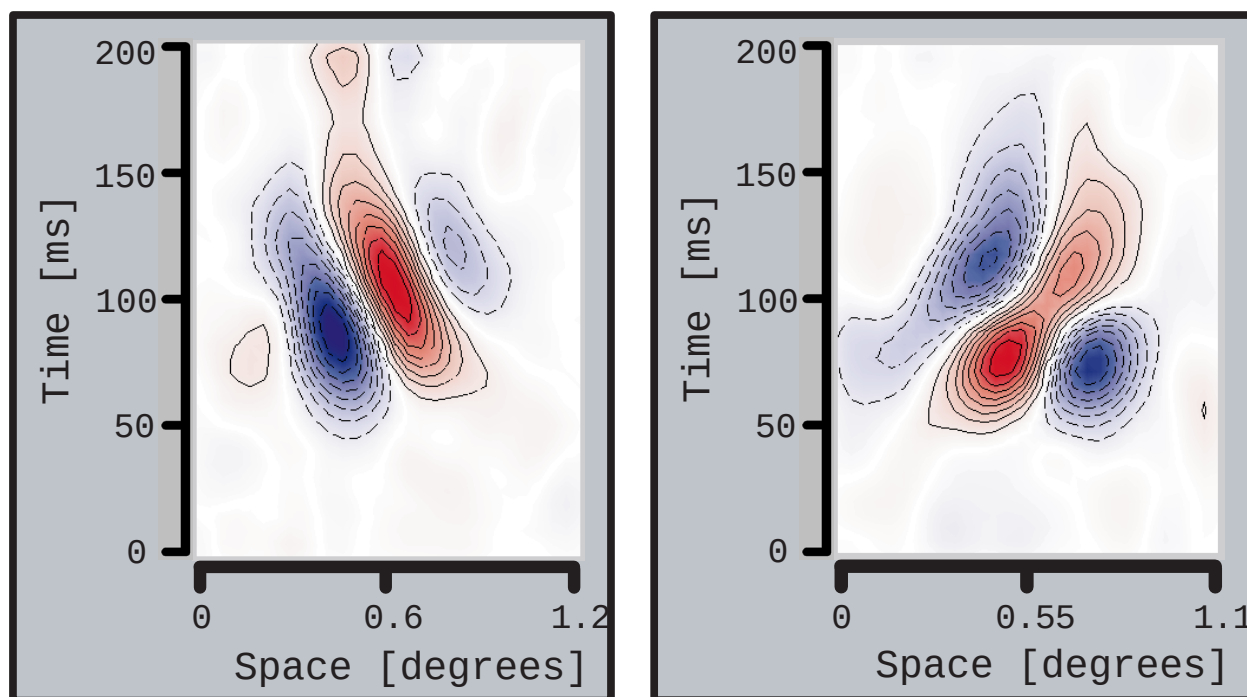


Fig. 6. Examples of the spatio-temporal receptive fields (RFs) of two directionally-selective V1 simple cells. Note that while the sub-regions of the RFs of the non-directional cells shown in Fig. 2 all remain in the same location over time, the sub-regions of the RFs of directional cells shift in space over time, producing oriented spatio-temporal RFs. Note, however, that the overall RF of these directional cells does not shift in position over time as one would expect if they were summing the outputs of two cells with similar RFs but in different spatial positions.

includes both 90 and 270°; 0° includes both 0 and 180°; etc.). It can be seen that about 80% of the fast biphasic cell population, but only about 20% of the slow monophasic cells, have even-symmetric (cosine) spatial RFs.

As an aside not relevant to the thrust of this paper, but nonetheless of interest, we consistently found that among those simple cells with cosine RFs, a greater number have RF centers responsive to decrements in luminance (i.e. approximately 180° spatial phase RFs), than have RF centers responsive to luminance increments (0° spatial phase RFs).

As can be seen from the data shown in Figs. 4 and 5, and as will be discussed further below, if a later V1 cell received inputs from random samples of each of these sub-populations of non-directional cells, the two sets of inputs would almost always be in temporal quadrature, and often would be in different spatial phases as well, thus providing the cell with some degree of direction selectivity. About 20% of the time, a random combination of the two samples would be in almost complete temporal and spatial quadrature and would produce a highly directional cell. It should also be clear that the biphasic cells within a cortical column by themselves have neither the requisite temporal nor the requisite spatial differences in their RFs to capture motion information.

3.6. Spatio-temporal RFs and latencies of directional cells

Using the criterion of a cell's having a Direction Index of more than 70, in response to a grating of optimal spatial frequency and orientation drifting in the preferred and non-preferred directions, 37 of the 171 striate cortex cells were identified as being directionally-selective. Fig. 6 shows the spatio-temporal RFs of two typical directional cells. It can be seen in each case that after an initial latency, the response builds up and then shifts in position over time. These cells thus have a RF which is oriented in space-time, just as one would expect from the linear motion models of Adelson and Bergen (1985) and Watson and Ahumada (1985). All of the directionally-selective cells in the sample had such space-time oriented RFs. As one would predict from such oriented spatio-temporal RFs, these directionally-selective cells respond optimally to a pattern moving in one particular direction, respond poorly to a stationary pattern, and respond almost not at all to a pattern moving in the opposite, non-preferred direction.

Directional cells typically have RFs made up of three or more strongly-antagonistic spatial regions, as can be seen in the examples shown in Fig. 6, and thus have bandpass spatial frequency tuning. The spatio-temporal RF mapping was only one dimensional in space, being

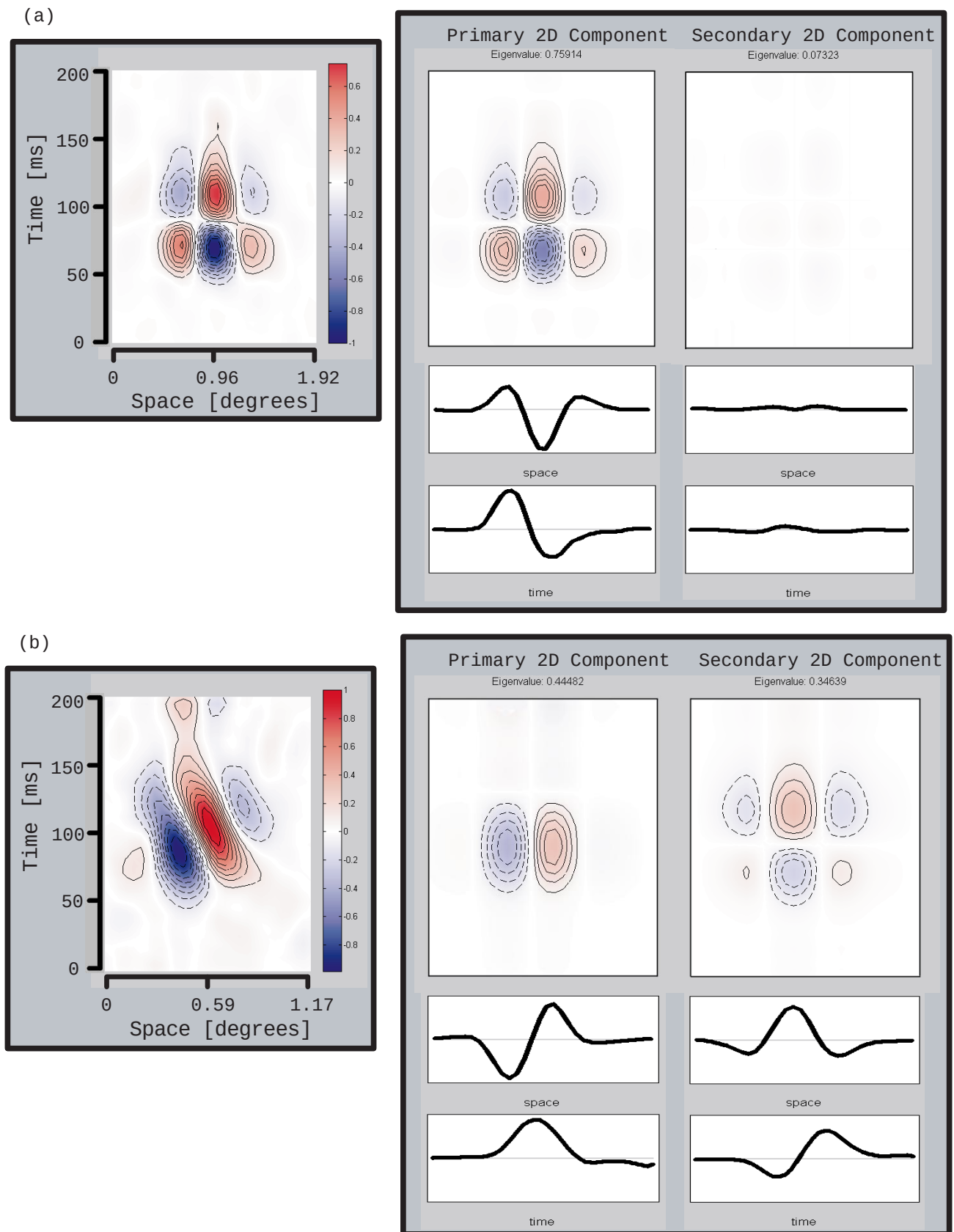


Fig. 7.

carried out with bars of the optimal orientation, and thus do not reveal the fact that the different RF sub-regions seen in cross-section in the data are also typically elongated in the orthogonal direction. However, tests with gratings show that the directional cells typically have bandpass orientation tuning, as well.

To analyze quantitatively the RF structure of striate cortex cells, a principal components analysis, or singular value decomposition (SVD), of the spatio-temporal RF of each of the cortical cells in the sample was carried out. This procedure computes the component that alone accounts for most of the RF shape, and the amount of each of various additional orthogonal components required to account for the entire spatio-temporal RF. Since the spatio-temporal RF of a typical biphasic or a monophasic non-directional cell does not change significantly in spatial position (or spatial phase) over time, the SVD shows that almost all of its RF structure typically can be accounted for by a single component. That is, a single spatial profile that is merely scaled proportionately (up or down) over time matches the profile of a typical non-directional cell's spatio-temporal RF. In Fig. 7a, as an example, the two principal components are shown that together respectively account for 76 and 7% of the structure in the RF of the non-directional cell shown in Fig. 2e. It is clear that the first SVD component captures almost all of the RF structure, the 2nd- (and higher-) order components being largely just noise.

The RF of a directional cell changes in space over time and thus a single component, a single spatial profile, cannot account for its spatio-temporal RF structure. However, it was found that the linear combination of just two components can account for almost all of the RFs of each of the directional cells in the sample. In Fig. 7b the two almost equally large SVD components are shown that account for 47 and 36%, respectively, of the RF structure of the directional cell shown in Fig. 6a. In this case, it is apparent that two strong SVD components are required to produce the spatio-temporally-oriented RF of this directional cell.

What is of interest is that the two principal components which, when combined linearly, can account for the RF structure of almost all directional cells, corre-

spond very closely in both temporal and spatial RF characteristics to the respective characteristics of the subpopulations of non-directional cells which have been identified, namely, the fast biphasic and the slow monophasic cells, respectively. That is to say, the spatio-temporal RFs of directional cells could be quite precisely accounted for if each were linearly summing inputs from a sample of fast biphasic cells and from a sample of slow monophasic non-directional cells in the same cortical column. So in the case of the cell whose RF is illustrated in Fig. 7b, the fastest component is strongly temporally-biphasic and is in cosine spatial phase, resembling in both respects the biphasic non-directional cell subpopulation. The slower SVD component required to construct the RF of the directional cell illustrated is monophasic and in sine phase, like many cells from the monophasic non-directional cell subpopulation.

The SVD analysis finds the various orthogonal components which when added together can produce the initial pattern. One might think, therefore, that if the 1st principal component of the SVD of a directional cell is in, say, cosine spatial phase that the 2nd component must by definition be in sine phase. That is not so. The two orthogonal components might each be in cosine phase (which, remember, one is defining with respect to the RF or component center), but be orthogonal by having one shifted in *position* by a quarter cycle with respect to the other, to put them in spatial quadrature.

One can also see by inspection of the spatio-temporal RF of the directional cell illustrated in Fig. 7b that while the separate RF bands shift in space over time, the overall RF does not change position over time. Rather, the initial cosine spatial profile changes in time to a sine profile, but the RF remains centered at the same location with the same total width. If directional cells were being constructed by summing two similar cells of magno LGN origin (each temporally-biphasic and spatially-triphasic) whose RFs were in *different* locations, one would expect the whole RF to shift to the left, which it does not do. Rather, the RFs of directional cells which have been recorded are typically centered on the same point across time, with the sub-bands but not the overall RF changing position,

Fig. 7. The results of singular value decomposition (SVD) analysis of the receptive fields (RFs) of two V1 simple cells. In each case, the cell's RF, and the two principal components of the SVD analysis of this RF are shown. Beneath each is shown the space and time slices which together can account for that SVD component. The space slice in each case is that constant spatial profile that fits every cross-section through the RF, when multiplied by the value at that time in the time slice. The time slice shows how the RF changes over time. (a) The results for the non-directional cell shown in Fig. 2E. It can be seen that one principal component accounts for almost all the variance in the RF (eigenvalue of 0.76, with an eigenvalue of 0.07 for the next largest component). This is what one would expect from a non-directional cell whose RF does not change in space over time. (b) The results for the directional cell shown in Fig. 6A. To account for this spatio-temporally oriented RF requires two almost equally large principal components (eigenvalues of 0.47 and 0.36). It can be seen that the fastest of these is very biphasic and in cosine spatial phase with respect to the RF center, and that the slower one is monophasic and in sine spatial phase. These two components thus correspond closely to the biphasic and monophasic non-directional cells populations that one postulated serve as the inputs to directional cells.

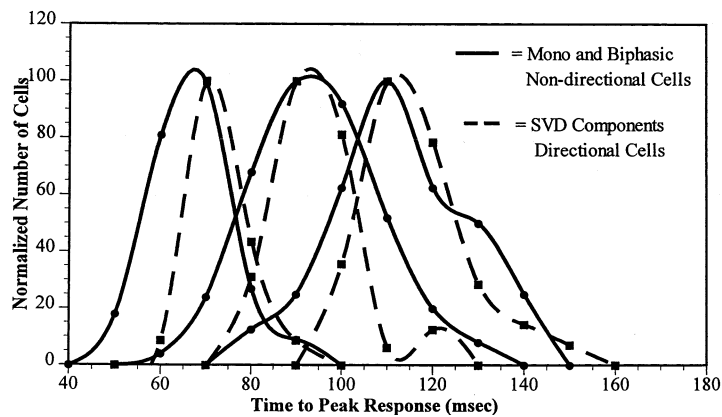


Fig. 8. A comparison of the distributions of time-to-peak response of the receptive fields (RFs) of the two non-directional cell populations (solid lines), and of the two principal singular value decomposition (SVD) components of the RFs of directional cells (dashed lines). It can be seen that these sets of distributions are virtually identical. The initial and reversed phases of the fastest SVD component of directional cells slightly lag in time the initial and reversed phases of the biphasic non-directional cells; the time-to-peak of the later SVD component of directional cells slightly lag in time the responses of the monophasic non-directional cells. This is what one would expect if directional cells combined inputs from monophasic and biphasic non-directional cells within a local region.

as one would expect if they were being built up by combining inputs of cells of different spatial and temporal *phases*, both samples being centered on the same spatial location.

In Fig. 8 are shown the times to peak response of the monophasic cells and both phases of the biphasic cells in comparison with the times to peak for the earliest SVD component of all the directional cells, the slower SVD component of these directional cells, and the reversed phase of their earliest component (which is quite biphasic). It can be seen that there is a remarkably close relation between the temporal properties of the different non-directional cell types and the temporal properties of the SVD components of the directional cells. The earliest (biphasic) SVD component of the directional cells reaches its initial and its reversed-phase peaks just shortly (about 4 ms) after the initial and the reversed-phase responses of the biphasic non-directional cells; and the second SVD component of the directional cells reaches its peak just shortly after the population of monophasic non-directional cells.

Furthermore, the spatial as well as the temporal characteristics of the principal components of the RFs of directional cells correspond to the two classes of non-directional cells which were postulated constitute their respective inputs. With respect to the RF center, the fast biphasic component in most directional cells is in cosine spatial phase and the later monophasic component is usually in sine spatial phase, just as would be expected if the initial input to directional cells came from the fast-biphasic, cosine-spatial-phase subpopulation, and the later, spatio-temporally-displaced input came from those many slow-monophasic cells with sine-spatial-phase RFs (see Fig. 9).

3.7. Temporal characteristics of parvo and magno LGN cells

The studies of the spatio-temporal RFs of directional and non-directional cells in the striate cortex led us to speculate (De Valois & Cottaris, 1998) that the two classes of non-directional cells that appear to be combined to form directional cells might have their origin in magno- and parvo-cellular LGN cells, respectively. To examine this further, the temporal (and spatial) properties of a population of cells in the P_c and M_c layers of the LGN were recorded, to compare with the recordings from cells in striate cortex.

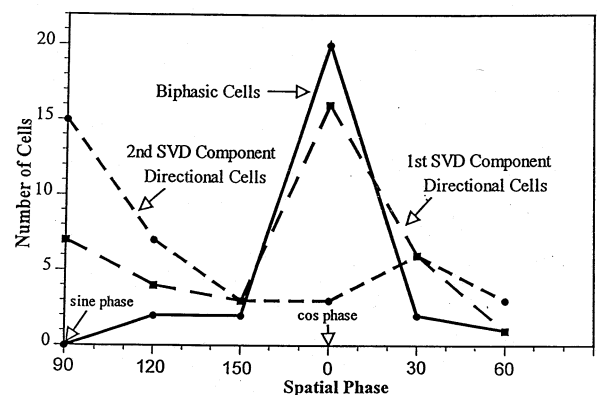


Fig. 9. A comparison of the spatial phases of biphasic non-directional cells and the singular value decomposition (SVD) components of directional cells. It can be seen that the fastest SVD components of directional cells are usually approximately in cosine spatial phase, as are the responses of the biphasic non-directional cells which one postulated constitute their fast inputs, and the slower SVD components of directional cells are at a variety of other spatial phases, as are the monophasic non-directional cells which one postulated as their slower inputs.

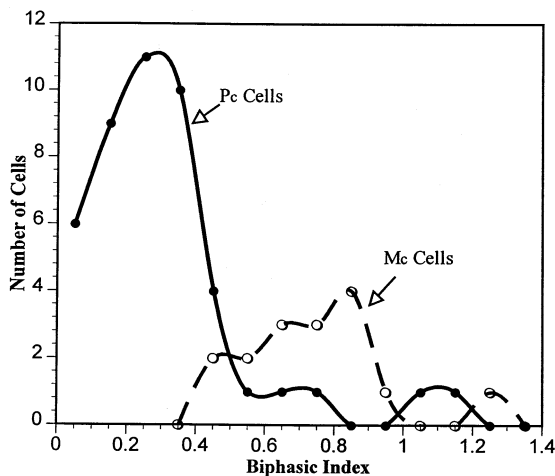


Fig. 10. The distributions of biphasic indices of parvocellular (P_c) and magnocellular (M_c) lateral geniculate cells. While lateral geniculate nucleus (LGN) cells in general are more biphasic than striate cortex cells (shown in Fig. 3), P_c cells, like monophasic non-directional V1 cells, are less biphasic than M_c (and the biphasic V1 non-directional sub-population) cells.

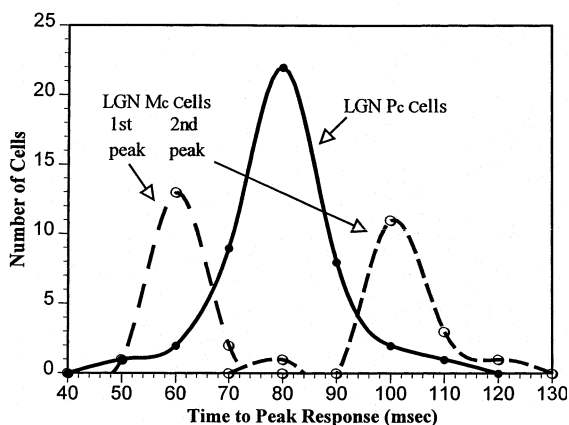


Fig. 11. The distributions of time-to-peak response in the receptive fields (RFs) of P_c and M_c lateral geniculate nucleus (LGN) cells. It can be seen that on average M_c cells have a shorter latency than P_c cells. Note that the responses of the P_c cells are going from one phase to the reverse. These two LGN cell types are thus in quadrature temporal phase.

A number of investigators have previously studied the temporal properties of LGN cells, but the precise latencies and phase characteristics of the responses of cells depend considerably on the stimuli used. Since we wanted to make direct, quantitative comparisons between the temporal properties of LGN and V1 cells, it was essential to examine the responses of LGN cells to exactly the same stimuli which was being used to study cells in the cortex.

Several previous investigators reported that P_c cells have a longer response latency than M_c cells, and have also described P_c cells as exhibiting sustained and M_c

cells transient temporal responses (e.g. Gouras, 1968, 1969; Marrocco, 1976; Schiller & Malpeli, 1977; Schmolesky et al., 1998). The spatio-temporal RF mapping results are consistent with these previous reports, and give further quantitative support for them.

The RFs of the LGN cells were determined using the same RF mapping procedures that were used in the cortical studies described above, namely, m-sequence reverse correlation with luminance increments and decrements. A total of 46 P_c and 16 M_c cells were studied. From the RF maps, the biphasic index and the latency to the response peaks were determined, as had been done with cortical cells.

In Fig. 10 are presented the distributions of the biphasic indices for the P_c and M_c cell populations. It can be seen that that M_c cells are in general much more biphasic than P_c cells. These quantitative data are consistent with the characterization (e.g. Schiller & Malpeli, 1977) of M_c cells as being transient and P_c cells as being sustained: the initial response of an M_c cell to a stimulus is sharply curtailed and falls into the opposite phase and thus is transient, whereas the more monophasic response of a P_c cell continues and only gradually fades away and thus is sustained.

It is not relevant to the main questions being addressed, but note, in comparing the data for cortical cells shown in Fig. 3 with those from LGN cells shown in Fig. 10, that LGN cells are on average more biphasic than cortical cells. Thus, P_c cells are more biphasic than monophasic V1 cells, and M_c cells are more biphasic than biphasic V1 cells. This suggests that the initial V1 circuitry acts to some extent as a low-pass temporal filter.

The authors also find, as have others (Schiller & Malpeli, 1977; Schmolesky et al., 1998), that M_c cells are faster in their responses than are P_c cells. The median latency to the start of the response, measured from the spatio-temporal RFs, was 38 ms for the M_c cells and 48 ms for the P_c cells (means of 37 and 51 ms, respectively). The absolute values and the 14 ms difference in means between the two populations are similar to those reported by Schmolesky et al. (1998). However, biphasic cells have in effect two responses to a stimulus, an initial phase and the reversed phase, and it is of more interest from the authors' point of view to look at the latency to the peaks of both phases of the response rather than to only the start of the initial response. Fig. 11 shows the distribution of peak response latencies for the P_c cells and time to peak for both the initial and the reversed-phase responses of the M_c cells. What is of interest in Fig. 11 is not just that parvo cells are slower than magno cells, but that the peak of the P_c cells' responses occurs right at the time that the responses of the M_c cells are going through zero as the response phase reverses. Thus they are in temporal quadrature with respect to each other, just as

are the two sub-populations of V1 cells which have been identified (see Fig. 4). Therefore, these two LGN cell populations potentially provide the striate cortex with exactly the temporal ingredients required for motion detection.

The populations of P_c and M_c cells differ not only in their latencies, but in the extent of temporal variations

within the populations as well. Magno cells are remarkably similar to each other in their latencies, with almost all showing initial response peaks at precisely 58 ms to the stimuli, whereas parvo cells show considerable variability in latency. Although P_c cells are in general slower, and some are very slow, others have as short a latency as M_c cells.

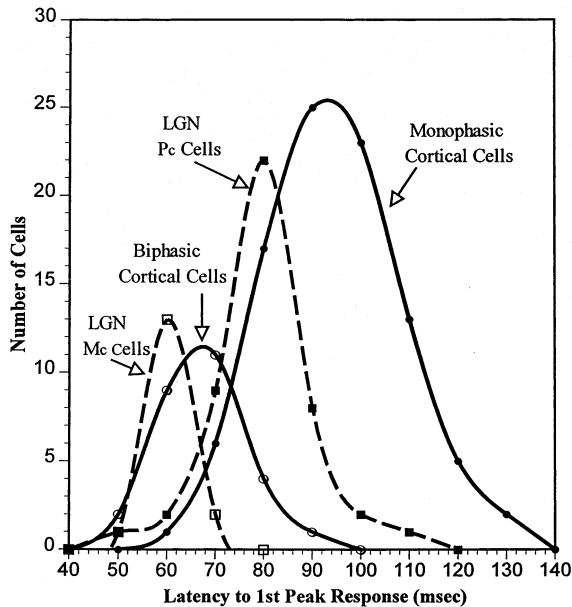


Fig. 12. A comparison of the times-to-peak response of the two lateral geniculate nucleus (LGN) cell populations (P_c and M_c cells), and the two V1 non-directional sub-populations which have been identified (monophasic and biphasic cells). The latency to peak response of the M_c cells is about 7 ms before what one postulated is their cortical target, the biphasic V1 cells. The latency to peak response of the P_c cells is about 12 ms before what one postulated is their cortical target, the monophasic V1 cells.

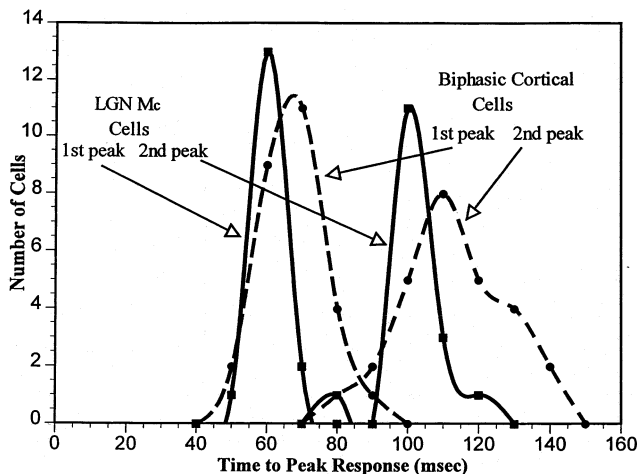


Fig. 13. A comparison of the times-to-peak response of both phases of the responses of M_c lateral geniculate nucleus (LGN) cells and biphasic V1 cells. It can be seen not only that the M_c cells' initial responses just slightly precede that of the biphasic cells, but that this is true of the later, reversed-phase responses as well.

3.8. Comparison of LGN and striate cortex cells

The primary reason for measuring the temporal characteristics of LGN cells was to examine whether or not the two cortical simple non-directional cell populations that have been identified, monophasic and biphasic cells, might have their origins in the parvo and magno LGN cells, respectively. Fig. 12 shows the peak latencies of the P_c and M_c LGN cells, and of the biphasic and monophasic striate cortex non-directional simple cells. It is apparent that the distribution of latencies of M_c LGN cells resembles and occurs slightly (about 7 ms) before that of biphasic V1 cells. The responses of P_c LGN cells show a similar distribution of time courses as the monophasic V1 cells, and precede them by about 12 ms in time. Note that the responses of the biphasic V1 cells precede those of parvo LGN cells, which thus could not be the source of their input.

That the fast biphasic non-directional V1 cells are likely getting their input from M_c cells, and the slow monophasic V1 cells from P_c cells, is also indicated by their respective laminar locations. Of those 23/27 fast biphasic cells whose laminar location could be specified, eight were found in layers 4b or 4c α , and 15 in lower layer 6, all layers receiving input from the magno LGN cells. None of the fast biphasic cells were found in the superficial layers. Slow monophasic cells, on the other hand, were found to be more widely distributed, in layers 2–3, in layer 4c β and in layers 5 and 6, all primarily related to the parvo LGN path. Novak, Munk, Girard, & Bullier (1995) examined not the phasic character but the latency of cells in various laminae and found that cells in layers 4b and 4c α had about 20 ms shorter latency than units in 4c β or in the supra- or infra-granular layers. These results are consistent with ours, and with the conclusion that the biphasic cells get their input from the M_c LGN path, and the monophasic cells from the P_c path.

Fig. 13 shows the latencies for both response phases of the M_c LGN cells and of the biphasic V1 cells. Clearly, the temporal distributions of the responses of biphasic cortical cells and magno LGN cells are very similar, not just for the initial, but for both response phases. It thus appears quite likely that the fast biphasic striate neurons receive their input from M_c LGN cells, and that the slow monophasic non-directional cells receive their input from P_c LGN cells.

4. Discussion

Although that issue was not the main focus of the investigation, it is important to note that the recordings from monkey striate cortex support the findings of Emerson & Gerstein (1977) and others from studies of cells in cat striate cortex in showing that separable spatio-temporal inputs combined to form spatio-temporally inseparable cells can account for the characteristics of directional cells in striate cortex.

This project had its origin when, in the course of RF mapping studies carried out for other purposes (Cotaris & De Valois, 1998), evidence was found for a bimodal distribution of striate cortex non-directional simple cells in terms of the biphasic nature of their temporal RFs. Most cells are largely monophasic in their temporal responses, but a distinct sub-population of V1 simple cells was found with very biphasic temporal responses. It was shown some time ago (Schiller & Malpeli, 1977) that P_c LGN cells have sustained and M_c cells transient temporal responses, a distinction that is consistent with P_c cells being monophasic and M_c cells being biphasic in their temporal responses. Since these two populations of LGN cells project to different sub-layers of the striate cortex, one might expect to find separate monophasic and biphasic striate cortex populations as well. Maunsell and Gibson (1992) looked for but did not find evidence for such a division, but m-sequence RF mapping with multiple stimuli produces significantly more robust RF maps than classical procedures. The m-sequence RF mapping data provide clear evidence for cells in V1, as well as in LGN, being divided into largely monophasic and very biphasic sub-populations. In fact, the data show that the distinction within the V1 simple cell population is even clearer than it is for P_c and M_c LGN cells.

Dividing the population of non-directional V1 simple cells purely on the basis of the extent to which they exhibit biphasic temporal properties, it was found that the biphasic cells have a shorter latency than the monophasic cells. The finding that particularly captured interest was that, as a consequence of the difference in latency between these two populations combined with their different biphasic characters, the temporal responses of these two sub-populations of V1 cells to RF mapping stimuli are almost exactly in temporal quadrature (90° out of phase with respect to each other). The monophasic cell population reaches its peak response just as the faster biphasic cell population is crossing through zero from one temporal phase to the reverse.

It was found that these two V1 sub-populations differ not only in their temporal properties, but in the spatial profiles of their RFs as well. Almost all the fast biphasic V1 simple cells have even-symmetric (cosine) RFs, whereas RFs of every spatial phase (with respect

to the RF center) are found about equally frequently among the monophasic cell population. Therefore, the combination of the outputs of fast-biphasic cells with outputs of some of the many slow-monophasic cells within the same region with odd-symmetric (sine) RFs would provide just the input required for a linear motion detector, such as those modeled by Adelson and Bergen (1985) and Watson and Ahumada (1985). In fact, the characteristics of the population of V1 directional cells that have been examined with the same RF mapping procedures shows that they could be constructed by linearly combining inputs not from cells in different locations, but from samples of each of the two non-directional cell populations which have been identified within a given region.

Furthermore, these two V1 sub-populations of non-directional cells resemble in their respective temporal properties the reported characteristics of P_c and M_c LGN cells, respectively, suggesting (De Valois & Cotaris, 1998) that the motion system may receive inputs originating not just from cells in the magno pathway, as widely postulated (e.g. Livingstone & Hubel, 1987; Maunsell, 1987), but rather from a combination of both magno and parvo LGN cells.

In the present study the spatio-temporal RFs of a large sample of V1 non-directional cells, of V1 directional cells, and of a population of cells in both P_c and M_c LGN layers have been examined, studying responses to exactly the same stimuli in each case. It is important to distinguish what we have directly established, with respect to the spatio-temporal RFs of cells in different LGN layers and among different types of V1 cells, from the inferences we would like to draw with respect to the inputs to the first stage of motion processing.

4.1. Findings

What the results clearly show are the following:

1. Parvo LGN cells have a longer initial response latency (by about 14 ms) than magno cells and reach their peak responses about 20 ms later than the initial response phase of the magno cells.
2. The peak responses of parvo cells occur on average at precisely the time that magno cells are crossing from one response phase to the other. These two classes of LGN cells are thus in temporal quadrature.
3. Although almost all LGN cells have a somewhat biphasic temporal response, magno cells are more strongly biphasic on average than parvo cells.
4. The distribution of the biphasic index (ratio of 2nd to 1st temporal response amplitudes) of all of the non-directional striate cortex simple cells is clearly bimodal, there being a small population of very biphasic cells, and a larger population of predominantly monophasic cells.

5. The V1 biphasic cells are primarily found in the magno-LGN-recipient regions of upper layer IV and lower layer VI.
6. Dividing the population of V1 cells solely on the basis of their biphasic indices, it was found that the biphasic cells are much faster than monophasic cells, reaching their peak (initial phase) responses about 25 ms earlier.
7. The monophasic population reaches its peak response just at the time at which the biphasic population is crossing over from one phase to the reverse phase. These two classes of non-directional simple V1 cells are thus in temporal quadrature.
8. Each of the biphasic response peaks of the magno cell population precedes those of the biphasic V1 cells by about 7 ms, and the peak responses of the parvo cell population precedes those of monophasic V1 cells by about 12 ms. However, the responses of biphasic V1 cells occur on average about 13 ms earlier than the responses of parvo LGN cells, which could thus not be their source.
9. Almost all biphasic V1 cells have RFs in even-symmetric (cosine) spatial whereas the spatial phases of monophasic V1 cells are scattered at all spatial phases. Therefore, most monophasic cells have RFs with different spatial phases from those of the biphasic cells, and some of the cells in these two subpopulations have RFs that are in, or near, spatial as well as temporal quadrature, with respect to each other.
10. Directionally-selective V1 simple cells have space-time oriented RFs. Principal components analysis of their RFs show that in most cases the RFs could be constructed by a linear combination of just two inputs: a fast temporally-biphasic input and a slower monophasic input.
11. The two inputs that can be summed linearly to produce the RFs of directional cells have the temporal and spatial characteristics of the sub-populations of non-directional cells which have been identified, namely the fast biphasic and the slow monophasic simple cells (and these in turn have the temporal and spatial characteristics of M_c and P_c LGN cells, respectively).

4.2. Inferences

Those are the findings, and from them the following conclusions are drawn:

1. The fast biphasic magno LGN cells form the inputs to the fast biphasic non-directional V1 simple cell population, and parvo LGN cells form the inputs to the slow monophasic non-directional V1 simple cell population.
2. The fast biphasic non-directional simple cells and the slow monophasic non-directional simple cells,

respectively, provide inputs in near temporal and spatial quadrature to produce directionally-selective simple cells. The extent to which these inputs would be in temporal quadrature would depend on the temporal frequency of the stimulus (which would be true for the classic model as well). Thus, the directional unit would have some degree of temporal frequency tuning.

3. The first stage of the motion system may thus be constructed not by some non-linear interaction, nor by summing the outputs of just magno-related cells in different cortical locations, but from a linear combination of the outputs of cells *within* a cortical column, cells which respectively have their origins in magno and parvo LGN cells.

It has been widely believed that the motion system consists of a pathway going from the magnocellular LGN layers to V1 layer IV α , to IVb, and from there directly and/or through V2 to MT and beyond. The study questions this, suggesting that it is not just cells in the magno LGN path that have input to the directionally-selective cells in V1, but that these directional cells are constructed from a combination of magno and parvo inputs. One must therefore reconsider the earlier evidence for the view that only magno cells form the input to the motion system.

One thing that has led to the view that the motion path has its origin just in the magno LGN cells is the early anatomical evidence that suggested not just a separate projection of P_c and M_c cells to different subregions of striate cortex layer IV, but separate pathways for these two systems throughout V1 and to later visual areas. However, there is increasing evidence for vertical interactions within cortical columns in V1 that would provide for a mixture of information originating in parvo and magno LGN layers (e.g. Lund & Boothe, 1975; Fitzpatrick, Lund, & Blasdel, 1985; Yoshioka, Levitt, & Lund, 1994), and indeed direct evidence for an input to layer IVb of the motion pathway from cells that have a P_c input (Sawatari & Callaway, 1996). Thus, the anatomical evidence no longer supports a complete segregation of these two paths, but rather indicates the possibility of interactions between them.

Anatomical evidence suggesting that the motion system receives input just from magno LGN cells also comes from the effects of LGN lesions on a monkey's ability to discriminate motion direction, and from the responses of cells in MT, which appears to be the primary motion-related cortical region. Both Schiller, Logothetis, and Charles (1990) and Merigan and Maunsell (1990) examined the effects of LGN lesions on discrimination performance in macaque monkeys and found major deficits in high temporal frequency motion and flicker discrimination following a magnocellular LGN lesion, but somewhat less loss to discrimination of low temporal frequency patterns.

Furthermore, Schiller et al. (1990) found little loss in motion discrimination after parvocellular LGN lesions. On the other hand, Maunsell, Nealey, and DePriest (1990) found losses in responsivity of motion-selective cells in MT after lesions to either P_c or M_c LGN laminae, although the effects of the magno lesions were greater. Furthermore, Merigan, Byrne, and Maunsell (1991) argue that M_c lesions (such as those used by Schiller et al., 1990) do not produce a specific deficit in motion perception but rather lead to a general loss in contrast sensitivity for high temporal frequency and low spatial frequency stimuli.

A third line of evidence which has been taken to indicate that the motion system has its origin just in the magno path is the observation (Shapley, Kaplan, & Soodak, 1981; Kaplan & Shapley, 1982) that magno LGN cells have a much higher contrast gain than do parvo cells, and saturate at quite low contrasts. Some psychophysical studies (e.g. Nakayama & Silverman, 1985) find that direction discrimination also saturates at very low contrasts, thus suggesting that motion analysis might be based on the outputs of magno cells. However, Edwards, Badcock, and Nishida (1996) find that performance in motion detection improves with increased contrast up to very high levels if the task is made more difficult. They argue that the low-contrast saturation seen in other motion studies is just a ceiling effect due to the simplicity of the particular task employed in those experiments.

One might also think that the evidence for a loss, or at least impairment, of motion at isoluminance (Ramachandran & Gregory, 1978; Livingstone & Hubel, 1987, 1988), might be in conflict with the suggestion of a major role for (color-coded) parvo cells in motion detection. However, in the model towards which the data are leading us, it is the combination of parvo and magno inputs which are critical. If the magno input were lost at isoluminance, the parvo input by itself would carry no directional information. The data thus do not conflict with the classic story in this regard.

On the other hand, several lines of evidence, in addition to the data present here, suggest a major role for parvo cells in motion perception. Anderson, Drasdo, and Thompson (1995) argue that the spatial sampling of the image by magno cells is too coarse to account for the spatial resolution of the motion system and thus that parvo cells must be involved. Also, a study of reversible lesions of different LGN laminae by Malpeli, Schiller, and Colby (1981) found that three of the four directional striate cells they examined lost their directional tuning after parvo lesions.

4.3. Phase versus position

The linear motion models of Watson & Ahumada (1985) and Adelson & Bergen (1985) postulate orthogo-

nal spatial and temporal inputs to the initial stage of direction selective units. These differing spatio-temporal inputs could either originate in units with the same spatial and temporal RF phase but shifted in position in space and time, respectively, or from units centered at the same location in space and time (so to speak) but in quadrature spatial and temporal RF phase. That is, the motion information could be based either on positional differences or on phase differences. From a purely theoretical point of view, these are arbitrary alternatives. Adelson & Bergen (1985), for instance, chose for their model two temporal inputs that are both biphasic but are shifted in time with respect to each other, but they point out that this was an arbitrary choice. Although these alternatives may be equivalent from a theoretical standpoint, they are not at all so with respect to the underlying physiology. The question raised is critical both with respect to the possible contributions of P_c and M_c LGN cells to motion and with respect to how directional cells might be constructed, given the columnar organization of the cortex. We have been emphasizing the issue of parvo and magno inputs, but now the problem posed by the cortical architecture is addressed.

One of the crucial advances of the linear motion models over the original Reichardt model is that they postulate inputs with spatial frequency and orientation selectivity, which would vastly simplify the correspondence problem. This is the problem, in a complex visual scene, of what in interval 2 and location 2 the motion system is to compare with what in interval 1 and location 1 to detect motion.

Most cells early in the cortical path are tuned for both spatial frequency and orientation. These cells could thus serve as the selective pre-filters to directional cells, to minimize the correspondence problem. However, there is a columnar organization of orientation (Hubel & Wiesel, 1962) and of spatial frequency (De Valois & De Valois, 1988), so that while cells in the same column are similarly tuned, those in neighboring columns have different spatial frequency and/or orientation tuning. The model which is being proposed, besides accounting for the data which has been presented, provides for a much simpler and more plausible mechanism for building a directionally-selective cell with selective orientation and spatial frequency tuning.

It was found that within a single cortical column are two non-directional cell types, the fast biphasic and slow monophasic cells (originating, we argue, from M_c and P_c LGN cells, respectively). Units of these two cell types within a column would have similar spatial frequency and orientation tuning, and many would have just the quadrature, or near quadrature, spatial and temporal RF phases, respectively, to produce directional cells, through a completely local circuit, with just linear summation of inputs.

One could of course with some ingenuity generate complex non-linear models with intra-cortical unidirectional delay lines by which one could produce a directional cell by combining fast biphasic cells (originating just from M_c LGN cells) in two different spatial locations in the cortex. However, there are at least three problems, two related to the results of this study and one related to the cortical architecture, that would have to be addressed by such an alternative, necessarily non-linear, model. One is that the SVD analysis of the directional cells shows that they can be linearly constructed from the combination of a biphasic and a monophasic temporal input, but not from two (M_c cell-like) temporally-displaced biphasic inputs. (This point is discussed at some length by De Valois & Cottaris, 1998). In an alternative model, some mechanism would have to be postulated to convert one of the inputs into a temporally monophasic shape. The same is true in the space domain: the SVD analyses of the directional cells show that they can be linearly constructed by a inputs respectively with cosine and sine RFs, but not from two cosine RFs spatially displaced with respect to each other. Again, some ad hoc mechanism would have to be postulated in an alternative model to handle this problem. Finally, because of the cortical architecture, the postulated lateral interactions would have to be specific distances apart because cells in neighboring cortical regions up to a millimeter away would all have differing spatial frequency and/or orientation tuning. A directional cell which received inputs via lateral connections from biphasic cells in near neighboring spatial locations would not have the bandpass orientation and spatial frequency selectivity found in directional cells. While such complex alternative models cannot be ruled out, the data show that they are not required. A much simpler model, involving just linear summation of the outputs of two classes of non-directional cells that differ in spatio-temporal phase, types of cells that were found to be present within a cortical column, can account for the characteristics of the directional cells reported in this study.

In many ways the extraction of motion information and of stereoscopic information involve the same computational problems. The question was raised (De Valois & De Valois, 1988) of whether stereopsis might be based on disparities between the RF phases of cells picking up from the two eyes rather than positional disparities in their RF locations, as had been classically postulated. Recent physiological studies of cells in cat cortex by Ohzawa, DeAngelis, and Freeman (1996) have given support to the position that interocular phase differences are indeed involved. The same issue was raised here with respect to motion.

References

- Adelson, E. H., & Bergen, J. R. (1985). Spatiotemporal energy models for the perception of motion. *Journal of the Optical Society of America A*, 2, 284–299.
- Albrecht, D. G., & Geisler, W. S. (1991). Motion selectivity and the contrast-response function of simple cells in the visual cortex. *Visual Neuroscience*, 7, 531–546.
- Albright, T. D. (1984). Direction and orientation selectivity of neurons in visual area MT of the macaque. *Journal of Neurophysiology*, 52, 1106–1130.
- Anderson, S. J., Drasdo, N., & Thompson, C. M. (1995). Parvocellular neurons limit motion acuity in human peripheral vision. *Proceedings of the Royal Society of London*, B261, 129–138.
- Anzai, A., Ohzawa, I., & Freeman, R. D. (1999). Neural mechanisms for encoding binocular disparity: position vs. place. *Journal of Neurophysiology*, 82, 874–890.
- Cottaris, N. P., & De Valois, R. L. (1998). Temporal dynamics of chromatic tuning in macaque primary visual cortex. *Nature*, 395, 896–900.
- de Boer, E., & Kuyper, P. (1968). Triggered correlation. *IEEE Transactions Biomedical Engineering*, 15, 169–179.
- De Valois, R. L., & Cottaris, N. P. (1998). Inputs to directionally selective simple cells in macaque striate cortex. *Proceedings of the National Academy of Sciences USA*, 95, 14488–14493.
- De Valois, R. L., & De Valois, K. K. (1988). *Spatial Vision*. New York: Oxford University Press.
- De Valois, R. L., Yund, E. W., & Hepler, N. (1982). The orientation and direction selectivity of cells in macaque visual cortex. *Vision Research*, 22, 531–544.
- Edwards, M., Badcock, D. R., & Nishida, S. (1996). Contrast sensitivity of the motion system. *Vision Research*, 36, 2411–2421.
- Emerson, R. C., & Gerstein, G. L. (1977). Simple striate neurons in the cat. II. Mechanisms underlying directional asymmetry and directional selectivity. *Journal of Neurophysiology*, 40, 136–155.
- Field, D. J., & Tolhurst, D. J. (1986). The structure and symmetry of simple-cell receptive-field profiles in the cat's visual cortex. *Proceedings of the Royal Society of London B*, 228, 379–400.
- Fitzpatrick, D., Lund, J. S., & Blasdel, G. G. (1985). Intrinsic connections of macaque striate cortex: afferent and efferent connections of lamina 4C. *Journal of Neuroscience*, 5, 3329–3349.
- Gouras, P. (1968). Identification of cone mechanisms in monkey ganglion cells. *Journal of Physiology (London)*, 199, 533–547.
- Gouras, P. (1969). Antidromic responses of orthodromically identified ganglion cells in the monkey retina. *Journal of Physiology (London)*, 204, 407–419.
- Hamilton, D. B., Albrecht, D. G., & Geisler, W. S. (1989). Visual cortical receptive fields in monkey and cat: spatial and temporal phase transfer function. *Vision Research*, 29, 1285–1308.
- Hartline, H. K. (1940). The receptive fields of optic nerve fibers. *American Journal of Physiology*, 130, 690–699.
- Hubel, D. H., & Wiesel, T. N. (1962). Receptive fields, binocular interaction and functional architecture in the cat's visual cortex. *Journal of Physiology (London)*, 160, 106–154.
- Jones, J. P., & Palmer, L. A. (1987). An evaluation of the two-dimensional Gabor filter model of simple receptive fields in cat striate cortex. *Journal of Neurophysiology*, 58, 1233–1258.
- Kaplan, E., & Shapley, R. M. (1982). X and Y cells in the lateral geniculate nucleus of macaque monkey. *Journal of Physiology (London)*, 330, 125–143.
- Kulikowski, J. J., & Bishop, P. O. (1981). Fourier analysis and spatial representation in the visual cortex. *Experientia*, 37, 160–162.
- Livingstone, M. S., & Hubel, D. H. (1987). Psychophysical evidence for separate channels for the perception of form, color, movement, and stereopsis. *Journal of Neuroscience*, 7, 3416–3468.

- Livingstone, M. S., & Hubel, D. H. (1988). Segregation of form, color, movement, and depth: anatomy, physiology and perception. *Science*, 240, 740–749.
- Lund, J. S., & Boothe, R. G. (1975). Interlaminar connections and pyramidal neuron organization in the visual cortex, area 17, of the monkey (*Macaca mulatta*). *Journal of Comparative Neurology*, 159, 5–14.
- MacLean, J., Raab, S., & Palmer, L. A. (1994). Contribution of linear mechanisms to the specification of local motion by simple cells in areas 17 and 18 of the cat. *Visual Neuroscience*, 1, 271–294.
- Malpeli, J. G., Schiller, P. H., & Colby, C. L. (1981). Response properties of simple cells in monkey striate cortex during reversible inactivation of individual lateral geniculate laminae. *Journal of Neuroscience*, 46, 1102–1119.
- Marrocco, R. T. (1976). Sustained and transient cells in monkey lateral geniculate nucleus: conduction velocities and response properties. *Journal of Neurophysiology*, 39, 340–353.
- Maunsell, J. H. R. (1987). Physiological evidence for two visual subsystems. In L. M. Vaina, *Matters of Intelligence* (pp. 59–87). Dordrecht: D. Reidel.
- Maunsell, J. H. R., & Gibson, J. R. (1992). Visual response latencies in striate cortex of the macaque monkey. *Journal of Neurophysiology*, 68, 1332–1344.
- Maunsell, J. H. R., Nealey, T. A., & DePriest, D. D. (1990). Magnocellular and parvocellular contributions to responses in the middle temporal visual area (MT) of the macaque monkey. *Journal of Neuroscience*, 10, 3323–3334.
- Merigan, W. H., & Maunsell, J. H. R. (1990). Macaque vision after magnocellular lateral geniculate lesions. *Visual Neuroscience*, 5, 347–352.
- Merigan, W. H., Byrne, C. E., & Maunsell, J. H. R. (1991). Does primate motion perception depend on the magnocellular pathway? *Journal of Neuroscience*, 11, 3422–3429.
- Movshon, J. A., Adelson, E. H., Gizzi, M. S., & Newsome, W. T. (1985). The analysis of moving visual patterns. In C. Chagas, R. Gatass, & C. Gross, *Pattern Recognition Mechanisms: Pontificiae Academiae Scientiarum Scripta Varia Faria* (pp. 117–151). Rome: Vatican Press.
- Nakayama, K., & Silverman, G. H. (1985). Detection and discrimination of sinusoidal grating displacements. *Journal of the Optical Society of America A*, 2, 267–274.
- Nealey, T. A., & Maunsell, J. H. R. (1994). Magnocellular and parvocellular contributions to the responses of neurons in the macaque striate cortex. *Journal of Neuroscience*, 14, 2069–2079.
- Novak, L. G., Munk, M. H. J., Girard, P., & Bullier, J. (1995). Visual latencies in areas V1 and V2 of the macaque monkey. *Visual Neuroscience*, 12, 371–384.
- Ohzawa, I., DeAngelis, G. C., & Freeman, R. D. (1996). Encoding of binocular disparity by simple cells in the cat's visual cortex. *Journal of Neurophysiology*, 75, 1179–1805.
- Ramachandran, V. S., & Gregory, R. L. (1978). Does colour provide an input to human motion perception? *Nature*, 2175, 55–56.
- Reid, R. C., Soodak, R. E., & Shapley, R. M. (1987). Linear mechanisms of directional selectivity in simple cells of cat striate cortex. *Proceedings of the National Academy of Sciences USA*, 84, 8740–8744.
- Reid, R. C., Victor, J. D., & Shapley, R. M. (1997). The use of m-sequences in the analysis of visual neurons: linear receptive field properties. *Visual Neuroscience*, 14, 1015–1027.
- Sawatari, A., & Callaway, E. M. (1996). Convergence of magno- and parvocellular pathways in layer 4B of macaque primary visual cortex. *Nature*, 380, 442–446.
- Schiller, P. H., & Malpeli, J. G. (1977). Properties and tectal projections of monkey retinal cells. *Journal of Neurophysiology*, 40, 428–445.
- Schiller, P. H., Logothetis, N. K., & Charles, E. R. (1990). Role of color-opponent and broad-band channels in vision. *Visual Neuroscience*, 5, 321–346.
- Schmolesky, M. T., Wang, Y., Hanes, D. P., Thompson, K. G., Leutgeb, S., Schall, J. D., & Leventhal, A. G. (1998). Signal timing across the macaque visual system. *Journal of Neurophysiology*, 79, 3272–3278.
- Shapley, R. M. (2000). The receptive fields of visual neurons. In K. De Valois, *The Handbook of Perception and Cognition*. San Diego, CA: Academic Press.
- Shapley, R. M., Kaplan, E., & Soodak, R. (1981). Spatial summation and contrast sensitivity of X and Y cells in the lateral geniculate nucleus of the macaque. *Nature*, 292, 543–545.
- Sutter, E. E. (1992). A deterministic approach to nonlinear systems analysis. In R. B. Pinter, & B. Nabet, *Nonlinear Vision* (pp. 171–220). Boca Raton: CRC Press.
- Tadmor, Y., & Tolhurst, D. J. (1989). The effect of threshold on the relationship between the receptive-field profile and the spatial-frequency tuning curve in simple cells of the cat's striate cortex. *Visual Neuroscience*, 3, 445–454.
- Watson, A. B., & Ahumada, A. J. (1985). Model of human visual-motion sensing. *Journal of the Optical Society of America A*, 2, 322–342.
- Webster, M. A., & De Valois, R. L. (1985). Relationship between spatial-frequency and orientation tuning of striate-cortex cells. *Journal of the Optical Society of America A*, 2, 1124–1132.
- Yoshioka, T., Levitt, J. B., & Lund, J. S. (1994). Independence and merger of thalamocortical channels within macaque monkey primary visual cortex: anatomy of interlaminar projections. *Visual Neuroscience*, 11, 467–489.
- Zeki, S. M. (1974). Functional organization of a visual area in the posterior bank of the superior temporal sulcus of the rhesus monkey. *Journal of Physiology (London)*, 277, 273–290.